VESCA SACRIFICAL ANODE

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Established : 17 Maret 1997
Trade Mark : VESCA

VESCA SKETCH

General
Services Activities

VESCA PRODUCT

Aluminium anode
Zinc anode
Magnesium

IMPRESSED CURRENT :

Rectifier
Platinum anode
Fe-Si anode
Graphite anode
Lead silver anode

VESCA SACRIFICAL ANODE

MATERIAL COMPOSITION
ALUMINIUM

Aluminium anodes are cast in alloy of aluminium, zinc and indium additions and have been development specifically for use in sea water and sea mud. The VESCA aluminium anodes are cast from the highest purity aluminium ingot to conform with the following material composition.

Material Composition And Technical Data

Element	Persentage (%)
Zinc	2.0 – 5.0
Indium	0.005 – 0.035
Titanium	0.01 – 0.05
Silicon	0.15 max
Iron	0.13 max
Copper	0.006 max
Others	0.02 max ea
Aluminium	remainder

Anode performance data

Nominal solution potential : -1.10 volt against Ag/AgCl
Nominal electrical capacity : 2700 Ampere hour per kg
Nominal comsumption : 3.24 Kg Amp/yrs
Nominal Driving voltage : 0.3 volts against polarized steel





VESCA SACRIFICAL ANODE

Aluminium

Aluminium anode for ship hull welding type

TYPE	Dimensi anoda (L x W x T) mm	Dimensi core (p x l x t) mm	Berat Nett (Kg)	Berat Gross (Kg)
AV-140	200 x 95 x 30	300 x 25 x 6	1.2	1.4
AV-320	250 x 145 x 30	350 x 40 x 6	2.6	3.2
AV-450	380 x 150 x 35	530 x 50 x 6	4.5	5.2
AV-550	380 x 150 x40	530 x50 x 6	5.5	6.0
AV-720	520 x 117 x 46	650 x 50 x 6	7.2	8.0
AV-860	520 x 117 x 52	650 x 50 x 6	8.6	9.8
AV-126	520 x 117 x 67	650 x 50 x 6	12.6	13.8
AV-141	915 x 130 x 50	1015 x 50 x 6	14.1	16.0
AV-210	915 x 130 x 75	1015 x 50 x 6	21.0	24.0
AV-300	920 x 130 x 140	1015 x 50 x 6	30.0	33.0



Aluminium anode for ship hull Bolt type

TYPE	Dimensi anoda (L x W x T) mm	Dimensi core (PCD)	Berat Nett (Kg)	Berat Gross (Kg)
B2	150 x 70 x 20	Bolts @ 75 mm between Ctrs	0.3	0.5
B3	200 x 100 x20	Bolts @ 110mm between Ctrs	0.8	1.0
B4	200 x 100 x 30	Bolts @ 110mm between Ctrs	1.15	1.4
B4E	200 x 100 x 40	Bolts @ 110mm between Ctrs	1.5	1.8
B8	300 x 150 x 25	Bolts @ 160mm between Ctrs	2.5	2.9
B9	300 x 150 x 30	Bolts @ 160mm between Ctrs	2.9	3.3
B9E	300 x 150 x 40	Bolts @ 160mm between Ctrs	4.2	4.6
B9F	300 x 150 x 50	Bolts @ 160mm between Ctrs	5.1	5.5
B11	300 x 150 x 75	Bolts @ 160mm between Ctrs	9.2	9.6

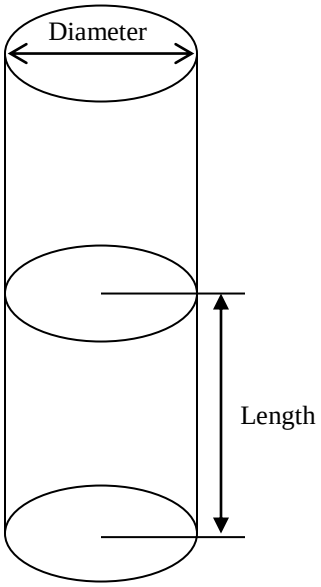


VESCA SACRIFICAL ANODE

Aluminium Rod Anodes

TYPE	Weight (Kg)	Length (mm)	Diameter (mm)
AVR-101	0.4	10	70
AVR-102	0.4	15	100
AVR-103	4.9	50	150
AVR-104	0.4	300	30
AVR-105	0.8	300	25
AVR-106	1.6	300	25
AVR-107	2.3	300	30
AVR-108	4.1	300	40
AVR-109	6.5	300	50
AVR-110	9.3	300	60
AVR-111	12.7	300	70
AVR-112	16.5	300	80
AVR-113	20.9	300	90
AVR-114	25.9	300	100
AVR-115	31.3	300	110
AVR-116	37.2	300	120
AVR-117	43.7	300	130



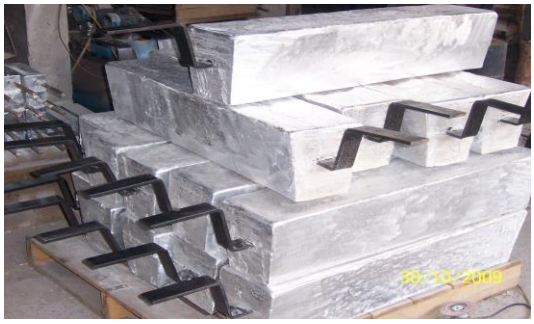


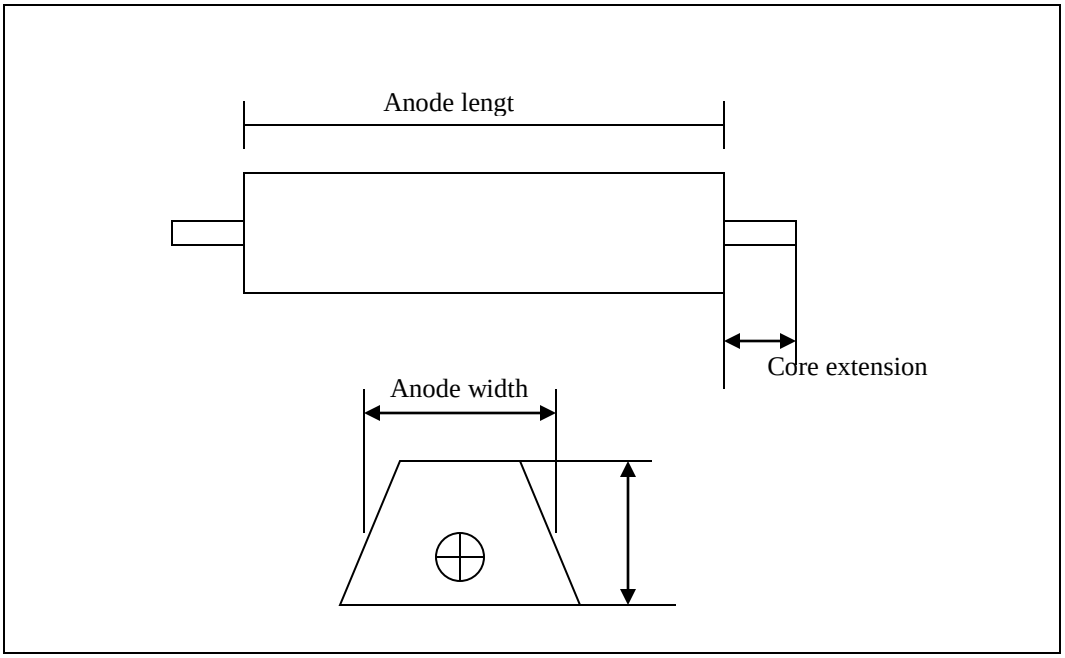
VESCA SACRIFICAL ANODE

PLATFORM OR JETTY
Aluminium Anodes

ANODE TYPE	Anode Length (mm)	Anode Weight (mm)	Anode Height (mm)	Core extension (mm)	Core diameter (mm)	Net Weight (Kg)	Gross Weight (Kg)
AVS-45	1320	102	102	152	19	35	38
AVS-68	1525	135	130	152	19	68	72
AVS-119	1550	184	159	152	25	119	126
AVS-148	2440	160	162	152	60	148	170
AVS-240	2750	207	207	152	114	240	306
AVS-330	2440	252	244	152	114	330	391
AVS-435	3000	254	257	152	114	435	507
AVS-565	3048	283	283	152	114	565	640







VESCA SACRIFICAL ANODE

MATERIAL COMPOSITION
ZINC

VESCA zinc andoes with the US Military Specification A-1800-J, the trade standard in general use worldwide. Our anodes are cast from 99.995% purity zinc alloy with small additions of special alloy element.

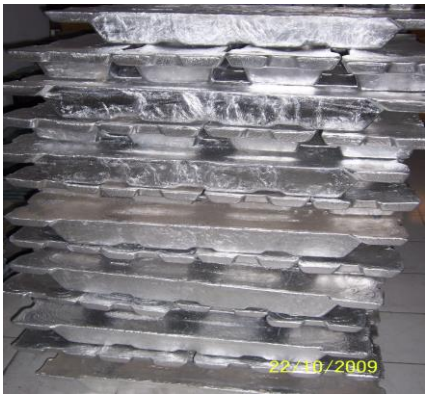
Material Composition And Technical Data

Element	A-1800-H	A-1800-J
Cadmium	0.025 – 0.15	0.025 – 0.07
Copper	0.005 max	0.005 max
Iron	0.005 max	0.005 max
Silicon	0.125 max	-
Aluminium	0.1 – 0.50	0.10 – 0.50
Lead	0.006 max	0.006 max
Zinc	remainder	remainder

Anode performance data

Nominal solution potential : -1.05 volt against Ag/AgCl
Nominal electrical capacity : 780 Ampere hour per kg
Nominal comsumption : 11.25 Kg Amp/yrs
Nominal Driving voltage : 0.25 volts against polarized steel





VESCA SACRIFICAL ANODE

Z I N C

Zinc anode for ship hull welding type

TYPE	Dimensi anoda (L x W x T) mm	Dimensi core (p x l x t) mm	Berat Nett (Kg)	Berat Gross (Kg)
S-2	150 x 70 x 20	230 x 24 x 3	1.87	2.0
S-3	200 x 100 x 20	300 x 38 x 3	2.60	3.0
S-4	200 x 100 x 30	300 x 38 x 3	3.80	4.2
S-6	300 x 150 x 20	270 x 30 x 4	5.75	6.0
S-8	300 x 150 x 25	270 x 30 x 4	7.25	8.0
S-9	300 x 150 x 30	270 x 30 x 4	8.75	9.5
ZV-320	270 x 150 x 32	350 x 40 x 6	5.5	6.0
ZV-450	400 x 150 x 32	530 x 40 x 6	11.1	12.1
ZV-165	550 x 150 x 25	650 x 40 x 6	16.5	18.0



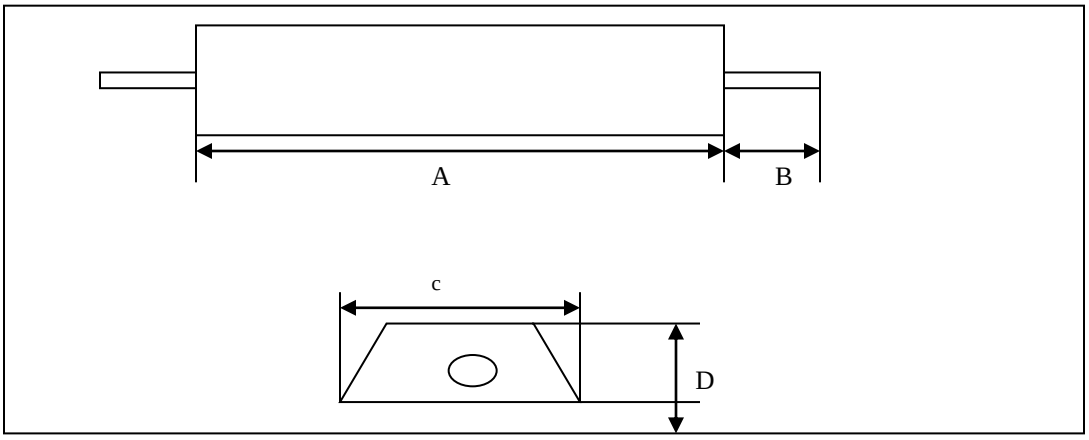
Zinc anode for ship hull B series

TYPE	Dimensi anoda (L x W x T) mm	Dimensi core (PCD)	Berat Nett (Kg)	Berat Gross (Kg)
B2	150 x 70 x 25	Bolts @ 75 mm between Ctrs	1.1	1.5
B3	200 x 100 x20	Bolts @ 110mm between Ctrs	2.2	2.5
B4	200 x 100 x 30	Bolts @ 110mm between Ctrs	3.3	3.6
B6	300 x 150 x 20	Bolts @ 160mm between Ctrs	5.5	5.75
B8	300 x 150 x 25	Bolts @ 160mm between Ctrs	6.9	7.3
B9	300 x 150 x 30	Bolts @ 160mm between Ctrs	8.2	8.5
B9E	300 x 150 x 40	Bolts @ 160mm between Ctrs	10.6	11.0
B9F	300 x 150 x 50	Bolts @ 160mm between Ctrs	13.4	13.8



VESCA SACRIFICAL ANODE

ZINC TANK ANODE









Anode Type	A mm	B mm	C mm	D mm	Core diameter mm	Net weight kg	Gross weight kg
ZVT-212	1220	227	51	51	12	21.2	22.7
ZVT-304	1220	227	60	60	12	30.4	32.0
ZVT-318	1220	227	63	63	12	31.8	33.4
ZVT-450	1220	227	76	76	12	45.0	46.6
ZVT-500	1525	227	70	70	12	50.0	52.0

VESCA

SACRIFICAL ANODE

Anodes ZINC ROD

TYPE	Weight (Kg)	Length (mm)	Diameter (mm)
AVR-101	1.1	10	70
AVR-102	3.3	15	100
AVR-103	12.5	50	150
AVR-104	1.0	150	30
AVR-105	2.1	300	25
AVR-106	4.2	300	25
AVR-107	6.0	300	30
AVR-108	10.7	300	40
AVR-109	16.7	300	50
AVR-110	24.1	300	60
AVR-111	32.7	300	70
AVR-112	42.8	300	80
AVR-113	54.1	300	90
AVR-114	66.8	300	100
AVR-115	80.8	300	110
AVR-116	96.2	300	120
AVR-117	112.9	300	130





PT. BIRO KLASIFIKASI INDONESIA (Persero)
CORROSION CENTRE
Address : Jl. Kacapiring No. 1 Bandung - 40271, INDONESIA
Phone : (62-022) 7275688, 7206854
Facsimile : (62-022) 7206854
E-mail : corrosion@cbn.net.id

CERTIFICATE OF ANALYSIS

DATE : 07.06.2007

Client : PT. Vescarindo Utama
Anode Alloy Material Spec : Zinc Anode
REF : Vesca-SA-P001

CHEMICAL ANALYSIS (ANODE LAB TEST)

El.	Zinc (Al)	Iron (Fe)	Lead (Pb)	Aluminum (Al)	Cadmium (Cd)	Copper (Cu)	Silicon (Si)
Ave.	97.8	0.0018	0.0023	0.1743	0.0018	0.0031	0.0027

Anode Capacity : 766 Ah/Kg
Open Circuit Potential : - 1.105 Volt vs Ag/AgCl
Consumption Rate : 0.476 Kg/ AY
Persen Effisiency : 90.8

QUALITY ASSURANCE ENGINEER

(Karyanto Herlambang)

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CERTIFICATE OF ANALYSIS

DATE : 07.06.2007

Client : PT. Vescarindo Utama
Anode Alloy Material Spec : Aluminum Anode
REF : Vesca-SA-P001

CHEMICAL ANALYSIS (ANODE LAB TEST)

El.	Aluminum (Al)	Zinc (Zn)	Magnesium (Mg)	Indium (In)	Iron (Fe)	Cadmium (Cd)	Copper (Cu)	Silicon (Si)
Ave.	97.8	2.7523	-	0.0187	0.0147	-	0.0312	0.1234

Anode Capacity : 2651 Ah/Kg
Open Circuit Potential : - 1.100 Volt vs Ag/AgCl
Consumption Rate : 0.138 Kg/ AY
Persen Effisiency : 88.9

QUALITY ASSURANCE ENGINEER

(Karyanto Herlambang)